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Modelling Crime Recurrence Timing at the Police Beat Level Using Survival Analysis and Deep Learning

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Declaration

I declare that this dissertation titled “Modelling Crime Recurrence Timing at the Police Beat Level Using Survival Analysis and Deep Learning” is my own original work and has been completed in accordance with the academic regulations of the University of Gloucestershire. All sources of information, ideas, data, and materials used in this work have been properly acknowledged and referenced.

I confirm that this dissertation has not been submitted, either in whole or in part, for any other degree or academic award.

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Abstract

Crime prediction often focuses on where crime happens or how many incidents may occur, but this does not directly show how soon another event may follow. This dissertation examines crime recurrence timing at the police Beat level, using robbery incidents in Chicago as the case study. The study transforms robbery records from 2001 to 2026 into survival intervals, where next_crime_gap measures the waiting time from one robbery to the next robbery in the same Beat. Cox Proportional Hazards, DeepSurv, LSTM-Cox, and a self-excitation-enhanced Cox model were compared using a time-aware train-validation-test split.

The results show that robbery recurrence is strongly concentrated in short time windows, with a median recurrence interval of 4.09 days. The self-excitation-enhanced Cox model achieved the best test C-index of 0.6251, while DeepSurv and LSTM-Cox did not outperform the best Cox-based model. Feature ablation showed that recurrence-history features produced the largest improvement, with neighbourhood activity and self-excitation adding further value. The study concludes that survival analysis can add useful timing insight to crime analysis, but the models should be treated as decision-support tools rather than exact prediction systems.

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CHAPTER 1: Introduction

1.1 Overview and Motivation

Crime remains a persistent challenge for cities worldwide, with consequences for public safety, economic activity, and community confidence (Institute for Economics and Peace, 2025). Despite continued investment in policing and prevention, crime does not occur in a regular or predictable way. Instead, incidents are unevenly distributed across both space and time. Some areas experience repeated events within short intervals, while others remain relatively inactive for extended periods before another incident occurs (Sherman, Gartin, and Buerger, 1989; Weisburd, 2015).

This variability is not simply descriptive. It has direct implications for decision-making. Interventions such as patrol deployment or targeted prevention measures are often time sensitive. Acting too early may waste resources, while acting too late may fail to prevent an event. As a result, understanding how crime evolves, and not just where it occurs, is important for practical planning.

In recent years, crime analysis has become increasingly data-driven, supported by the availability of large-scale police records and open data initiatives (Perry et al., 2013). The transition to geocoded and timestamped incident data has enabled more advanced analytical approaches. Statistical and machine learning models can now identify spatial clusters, detect temporal trends, and forecast the number of incidents expected within defined time windows (Kounadi et al., 2020; National Academies of Sciences, Engineering, and Medicine, 2025).

Despite these advances, most existing approaches continue to focus on two closely related questions: where crime is concentrated and how many incidents are likely to occur. While useful, these perspectives do not directly address the timing of recurrence. In practice, decision-makers require more than an understanding of long-term risk levels. They also need to identify areas that are approaching another event and may require immediate attention. Methods based on hotspot mapping or count prediction provide only indirect signals for this purpose and can be difficult to interpret in settings where crime occurs irregularly, with long quiet periods followed by sudden recurrence (Perry et al., 2013; Kounadi et al., 2020).

This study focuses on that timing dimension. Rather than modelling crime as counts or spatial density, it treats crime as a sequence of events and examines the time between consecutive incidents within the same area. In this setting, the police beat is considered as the unit of analysis, and the key question becomes how long a beat is likely to remain without another recorded incident. This framing shifts the focus from overall risk to recurrence timing, allowing the analysis to capture patterns that are not visible through standard approaches.

To address this problem, the study applies survival modelling techniques to robbery recurrence at the police Beat level. A Cox Proportional Hazards model is used as an interpretable baseline because of its established role in time-to-event analysis (Cox, 1972). This is extended through DeepSurv to examine nonlinear relationships, LSTM-Cox to model sequential event history, and a self-excitation-enhanced Cox model to test whether decayed same-Beat event history improves recurrence-risk ranking. The objective is therefore not only to model robbery recurrence timing, but also to assess whether predictive performance is improved more by model flexibility or by explicitly representing recent recurrence patterns.

1.2 Problem Statement

The central problem addressed in this study is that existing crime analysis approaches can identify where robberies are concentrated, but they do not directly model when the next robbery is likely to occur within the same local area. While hotspot maps and count-based forecasts effectively show high-crime locations, they provide limited insight into the waiting time until the next robbery event at the police Beat level. This creates a gap in understanding robbery as a recurrent timing process.

This gap matters because effective crime prevention and police resource allocation are highly time-sensitive. Knowing that a Beat has a historically high robbery rate is useful, but it does not indicate whether that Beat is currently at elevated risk of near-term recurrence. As a result, interventions may miss critical prevention windows.

The issue directly affects police planners, crime analysts, and communities experiencing repeated robbery incidents. Without modelling recurrence timing, resources may be allocated based mainly on long-term crime concentrations rather than current recurrence pressure. This can lead to inefficient deployment, missed opportunities for prevention, and limited understanding of how recent robbery history influences future risk.

At the academic level, although near-repeat victimisation and self-excitation studies suggest that recent events may temporarily elevate the risk of further incidents, this phenomenon has rarely been examined as a Beat-level survival analysis problem focused on time to next event. This study addresses this gap by modelling the timing of robbery recurrence within police Beats and examining whether recent event history, neighbourhood activity, and self-exciting patterns help explain shorter waiting times between robberies.

1.3 Research Questions

The study is guided by the following research questions:

- RQ1: What distributional patterns characterise robbery recurrence intervals at the police Beat level?
- RQ2: How well can survival models rank robbery events by recurrence timing under a time-aware evaluation framework?
- RQ3: To what extent do different feature groups, including recurrence history, neighbourhood activity, seasonality, and self-excitation, improve recurrence-risk ranking?
- RQ4: Do more flexible survival models, including DeepSurv and LSTM-Cox, provide meaningful improvements over an interpretable Cox Proportional Hazards baseline?
- RQ5: Which temporal and spatial features are most strongly associated with shorter robbery recurrence intervals?

1.4 Research Objectives

The aim of this study is to develop and evaluate models that capture the timing of crime recurrence at the area level. This is achieved through the following objectives:

- RO1: Construct an interval-level survival dataset from raw robbery incident records, representing the waiting time from a current robbery event to the next robbery within the same police Beat.
- RO2: Examine the distributional characteristics of robbery recurrence intervals at the Beat level.
- RO3: Develop temporally valid feature groups capturing recurrence history, neighbourhood activity, seasonality, and self-excitation, ensuring predictors are based only on information available at the prediction point.
- RO4: Fit a Cox Proportional Hazards model as an interpretable baseline for recurrence-risk ranking and hazard ratio interpretation.
- RO5: Implement and compare DeepSurv, LSTM-Cox, and self-excitation-enhanced Cox models to assess whether nonlinear, sequential, or event-history-based approaches improve predictive performance.

- RO6: Evaluate all models using a time-aware train–validation–test split, with performance assessed using the concordance index (C-index).
- RO7: Analyse the contribution of different feature groups and identify the temporal and spatial predictors most strongly associated with shorter recurrence intervals.

1.5 Ethical Considerations

Recorded crime data reflect officially reported incidents and may contain biases linked to policing practices, patrol intensity, and differential reporting behaviour (Lum and Isaac, 2016; Richardson et al., 2019). Therefore, the model outputs are interpreted as patterns in recorded crime rather than complete representations of all robbery activity. To minimise ethical risk, the study was conducted at the aggregated police Beat level and did not involve individual-level prediction, profiling, or identification. The analysis is intended for academic understanding of robbery recurrence patterns, not for real-time enforcement or individual targeting.

The dataset was obtained from the City of Chicago's public data portal and contains no direct personal identifiers. An institutional ethics waiver was granted by the university because the study used publicly available, anonymised secondary data and did not involve direct human participation.

Chapter 2: Literature Review

This chapter reviews the theoretical and methodological literature needed to frame crime recurrence timing at the police Beat level. It covers place-based crime theory, near-repeat victimization, survival analysis, self-exciting point processes, spatio-temporal crime prediction, and deep survival modelling. The review is organised to move from the foundational theoretical arguments that justify a location-level recurrence framework, through the methodological traditions of survival modelling and self-exciting process models, and into the broader landscape of modern predictive approaches. The chapter evaluates what each can and cannot explain, building toward a synthesis that identifies the specific gap this study addresses.

2.1 Theoretical Foundations: Crime, Place, and Temporal Dynamics

The conceptual basis for treating geographic locations as units of analysis with meaningful temporal histories is well established in criminological theory. Routine Activity Theory, as originally formulated by Cohen and Felson (1979), argued that criminal events require the convergence in time and space of three elements: a motivated offender, a suitable target, and the absence of a capable guardian. This framework shifted analytical attention away from offender characteristics toward the situational conditions that enable crime, laying the foundation for later place-based approaches that treated locations and their recurring activity patterns as important analytical units. The theory does not merely describe static risk but implies temporal dynamics: as the routine activities that structure daily life create predictable windows of opportunity, crime risk at a given location fluctuates in ways that are at least partially foreseeable.

Brantingham and Brantingham (1984) extended this line of reasoning through Crime Pattern Theory, which situates offending within the broader movement patterns of both offenders and victims through urban space. In this framework, crime concentrates not randomly but along activity nodes, pathways, and the edges between different land-use zones, producing structural regularities that persist over time. These theoretical frameworks provide the conceptual grounding for the empirical observation, now extensively documented, that crime is systematically concentrated at specific places. Sherman, Gartin, and Buerger (1989), in an influential analysis of calls for service in Minneapolis, found that approximately three percent of street addresses generated over half of all police calls, establishing what has since become known as the hot spots phenomenon. Weisburd (2015), reviewing decades of subsequent evidence across multiple cities and crime types, articulated this as the *"law of crime concentration"*, arguing that the regularity of geographic concentration is as reliable an empirical finding as exists in criminology.

The significance of these findings for the present study lies not merely in confirming that some places are riskier than others, but in the temporal dimension implied by persistent concentration. If certain locations consistently generate elevated crime rates across extended observation periods, this implies a degree of temporal dependency in the crime process at those locations: the history of a place is informative about its future. Weisburd et al. (2004) reinforced this point through trajectory analysis of crime at street segments over fourteen years in Seattle, finding that most high-crime locations maintain elevated rates with considerable stability, while a smaller subset exhibit meaningful change.

However, while these theories and empirical findings establish that crime is spatially concentrated and often stable over time, they offer less clarity on how risk evolves between events. In particular, they do not provide a framework for modelling the timing of recurrence at a given location, beyond suggesting that past activity is informative.

2.1.1 Near-Repeat Victimisation and Temporal Decay

A more fine-grained articulation of temporal dependence at the location level comes from the near-repeat victimisation literature. Building on earlier work on repeat victimisation (Farrell and Pease, 1993), this body of research demonstrates that a crime event elevates the risk of a subsequent event at or near the same location for a period of time thereafter, before risk decays back toward baseline. Johnson (2008) provided one of the most rigorous early examinations of this pattern in the context of residential burglary, showing that risk is significantly elevated in the immediate aftermath of an event and declines rapidly over time in a manner consistent with communicative or contagion-based processes, either through repeat offending by the same offender or through the diffusion of information within offender networks. The strongest elevated risk was concentrated within the first few weeks following an event.

Townsley, Homel, and Chaseling (2003) demonstrated similar patterns in an Australian context, and crucially showed that the near-repeat effect is strongest for events of the same crime type, lending support to a contagion interpretation rather than a purely environmental one. Piza et al. (2017) extended this literature through predictive modelling of initiator and near-repeat events across multiple offence categories, demonstrating that recent event history can improve short-term forecasting performance. The near-repeat paradigm has continued to develop: Rosenfeld, Vogel, and McCord (2019) examined near-repeat patterns in gun violence, finding evidence for clustering at finer temporal and spatial scales than those observed in property crime, suggesting that the decay dynamics vary meaningfully by crime type.

Despite these advances, the near-repeat literature remains largely descriptive: it characterises the decay of elevated risk in aggregate terms but does not provide a formal time-to-event model for estimating the waiting time until the next event at a specific location. This highlights the gap between characterising risk patterns and operationalising them as predictive timing models, which is a central motivation for the present study.
This temporal concentration of risk maps naturally onto the hazard function in survival analysis, where recurrence risk is elevated immediately after an event and declines as the event-free interval increases.

It is also worth noting a point of theoretical disagreement within the near-repeat literature. While the contagion interpretation has dominated, an alternative view holds that near-repeat clustering reflects shared environmental vulnerability rather than genuine event dependence: nearby locations share the same risk factors and therefore tend to be victimised at similar times without any causal link between individual events (Bernasco, 2008). This distinction is analytically important because it determines whether recurrence risk should primarily be modelled as event-driven temporal dependence or as the persistence of underlying environmental risk factors.

2.2 Survival Analysis: Framework and Applications

Survival analysis is relevant to this study because the research problem is concerned with timing rather than simple occurrence. Conventional crime prediction often asks whether crime will occur, where it is concentrated, or how many incidents may happen within a fixed period. Survival analysis shifts the focus to the waiting time until the next event. This is important for robbery recurrence because two Beats may have similar total robbery counts but very different recurrence patterns: one may experience repeated events within days, while another may have longer intervals between incidents.
The Cox Proportional Hazards model is widely used because it avoids imposing a fixed distribution on event times while still allowing covariate effects to be interpreted through hazard ratios (Cox, 1972; Kalbfleisch and Prentice, 2002). This is useful for crime recurrence

because the duration between robberies is unlikely to follow a simple parametric form. Robbery recurrence may be rapid in some Beats, irregular in others, and affected by both recent event history and local context. A semi-parametric survival model is therefore a defensible starting point.

However, the Cox model also introduces assumptions that must be treated critically. Its proportional hazards assumption implies that covariate effects remain constant over time. This is potentially problematic for crime recurrence because the influence of a recent robbery is likely to weaken as time passes. Stensrud and Hernán (2020) argue that proportional hazards should not be accepted uncritically, as violations are common in applied settings. In the context of near-repeat crime, this concern is especially relevant: if recurrence pressure decays after an event, then the effect of recent history may not be constant across the full duration period.

For this reason, Cox is best understood as an interpretable baseline rather than a complete theory of robbery recurrence. Its value lies in providing a transparent framework for modelling time to next robbery and interpreting which features are associated with shorter or longer recurrence intervals. Its limitation is that time-varying and event-driven dependence must be represented carefully through feature construction or alternative model extensions.

A further issue is that crime at a location is not a single-event process. A police Beat can experience repeated robberies across the study period, meaning the same spatial unit repeatedly returns to risk after each event. Standard survival models were originally designed for time to a first event, so applying them to robbery recurrence requires a recurrent-event perspective.

The Andersen-Gill model extends the Cox framework to recurrent events using a counting-process formulation (Andersen and Gill, 1982). This provides a rigorous way to model repeated event intensity over time. Frailty models offer another extension by allowing unobserved heterogeneity between units, which is useful where some units are persistently more active than others for reasons not captured by measured covariates (Box-Steffensmeier and Jones, 2004). These approaches are statistically attractive because they recognise that repeated events within the same unit are not independent in a simple sense.

Nevertheless, these models are not automatically the best fit for the present research objective. Counting-process recurrent-event models are often designed to estimate event intensity over time, whereas this study focuses on constructing directly interpretable recurrence intervals and comparing how well different models distinguish shorter from longer waiting times. The objective is not only to estimate an event process, but to model the waiting time from a current robbery to the next robbery within the same Beat. An inter-event survival formulation, therefore, provides a clearer alignment with the research question.

This choice still requires strict temporal discipline. Time-varying features must be constructed using only information available before the prediction point. Suresh et al. (2022) emphasise that survival models using longitudinal information are vulnerable to leakage when future information is allowed to enter predictor construction. This is particularly important for crime recurrence, where lagged gaps, rolling features, and neighbourhood activity could easily become invalid if they are calculated using future events. The value of a recurrent-event formulation therefore depends not only on the survival model selected, but also on whether the event history is represented without temporal leakage.

2.2.1 Applications in Criminology

Survival analysis has been widely used in criminology, but mainly for individual-level outcomes such as time to reoffending, rearrest, or reconviction. Yukhnenko et al. (2019) show that survival methods are valuable in recidivism research because they retain timing information that would be lost in binary reoffending outcomes. More recent work, such as Hamilton et al. (2023), also demonstrates the usefulness of survival models for examining how changes in individual circumstances affect reoffending hazards over time.

These applications demonstrate the criminological value of survival analysis, but they do not directly solve the problem addressed in this dissertation. In recidivism research, the unit at risk is usually an individual offender. In this study, the unit at risk is a geographic area. This changes both the interpretation of the event and the meaning of the predictors. The event is not an individual reoffending, but the recurrence of robbery within a police Beat. The predictors are not personal characteristics, but temporal history, neighbourhood activity, and local recurrence pressure.

Some place-based crime methods already incorporate temporal decay logic. For example, prospective hotspot mapping uses recent incidents to estimate short-term spatial risk (Bowers and Johnson, 2004a). However, such approaches usually remain closer to hotspot prediction than formal time-to-event modelling. They identify areas of elevated risk but do not directly model the waiting time until the next event. This leaves a methodological gap between place-based crime prediction and survival analysis.

The present study sits within that gap. It uses survival analysis to model robbery recurrence as an interval-level process at the police Beat level. This allows the analysis to retain the timing information contained in repeated robbery events while also supporting interpretable comparison of temporal, spatial, and recurrence-history features. The contribution is therefore not simply the use of survival analysis, but its adaptation from individual-level criminological outcomes to area-level robbery recurrence timing.

2.3 Self-Exciting Point Processes

The closest existing methodological tradition to the framework proposed in this study is the self-exciting point process literature, particularly Hawkes process models applied to crime. A Hawkes process (Hawkes, 1971) is a temporal point process in which each event triggers a temporary increase in the rate of subsequent events, with the triggering effect decaying over time. This temporal self-excitation formalises closely the near-repeat dynamics discussed in Section 2.1.1 and has been widely interpreted as a mathematical representation of crime contagion processes (Mohler et al., 2011; Short et al., 2008).

Mohler et al. (2011) introduced Hawkes process models to crime prediction, demonstrating substantial improvements over kernel density estimation in forecasting future burglary locations. The model decomposes crime occurrence into two interacting components: a background intensity representing long-run structural risk, and a triggered component representing elevated short-term risk following recent events. This distinction aligns closely with the broader criminological debate between environmental vulnerability and event-driven clustering (Bernasco, 2008; Johnson, 2008). In this sense, Hawkes models provide a formal statistical framework capable of representing both persistent place-based risk and short-term contagion effects within a unified process.

Subsequent research extended these models into operational policing contexts. Mohler et al. (2015) applied a Hawkes-based predictive policing system in Los Angeles, reporting reductions in crime relative to analyst-directed patrol deployment. However, these applications also attracted significant criticism concerning transparency, accountability, and the potential reinforcement of historical policing biases (Lum and Isaac, 2016). These criticisms do not invalidate the predictive utility of self-exciting models, but they highlight the importance of interpretability and careful evaluation when predictive systems are applied in operational decision-making environments.

The self-exciting point process framework has continued to evolve rapidly. Zhuang and Mateu (2019) extended Hawkes-type models by incorporating spatially heterogeneous triggering structures, allowing the strength and decay of excitation effects to vary across environments rather than remaining globally fixed. More recently, Mei and Eisner (2017) proposed a neural Hawkes process architecture that learns excitation dynamics directly from sequential event data rather than relying entirely on predefined parametric kernels. These developments reflect a broader methodological trend toward increasingly flexible representations of spatio-temporal dependence, particularly in settings where the functional form of the excitation process is not known in advance (Reinhart, 2018).

Despite these advances, several limitations remain relevant to the present study. Hawkes models are primarily formulated to estimate event intensity over continuous space-time, treating crime as a global stochastic process rather than as a set of discrete geographic units at risk (Daley and Vere-Jones, 2003; Reinhart, 2018). Although conditional intensity functions can theoretically be evaluated locally, model outputs are not naturally expressed as directly interpretable duration estimates or comparable recurrence rankings across predefined administrative units such as police beats. This distinction is operationally important because policing decisions are often made at the level of administrative areas, where the relevant question is not simply the intensity of crime events overall, but how soon another event is likely to occur within a specific unit.

From an operational perspective, the key issue is not only where robbery has historically concentrated, but whether recent event history suggests that a Beat is entering a shorter recurrence window. Standard Hawkes formulations are not usually expressed as Beat-level inter-event survival models, making them less directly aligned with the waiting-time outcome used in this study. In contrast, survival analysis produces this quantity naturally through the hazard and survival functions (Kalbfleisch and Prentice, 2002). From this perspective, survival models provide a more direct mapping between model outputs and decision-relevant operational questions.

Furthermore, standard and neural Hawkes models remain strongly dependent on event history itself and offer limited flexibility for incorporating rich sets of contextual and time-varying covariates without substantial increases in model complexity (Mei and Eisner, 2017; Reinhart, 2018). Survival modelling frameworks, by comparison, readily accommodate temporal, seasonal, and contextual predictors alongside event history, enabling the integration of recent activity patterns, spatial context, and temporal dynamics within a unified analytical structure. This flexibility is particularly important in crime settings where recurrence risk may depend not only on previous incidents but also on broader environmental and temporal conditions.

2.4 Spatio-Temporal Crime Prediction

2.4.1 Traditional and Intermediate Approaches

Kernel Density Estimation (KDE) has been the dominant tool for operational hotspot mapping, producing continuous risk surfaces from historical incident data (Chainey and Ratcliffe, 2005).

Prospective hotspot methods (Bowers et al., 2004b) address some of KDE's limitations by weighting recent events more heavily and applying decay functions that mirror near-repeat temporal patterns. Risk Terrain Modelling (RTM), developed by Caplan and Kennedy (2010), takes a different approach by identifying the environmental features, proximity to bars, transit stops, and abandoned buildings that together create criminogenic conditions, producing stable feature-based risk maps useful for strategic planning. Kennedy et al. (2011) demonstrated RTM's competitive predictive accuracy relative to density-based approaches, but the method's static character means it cannot capture the timing signal in recent event history, which is the focus of the present study.

2.4.2 Deep Learning Approaches

The past decade has seen a substantial influx of deep learning approaches to crime prediction. Convolutional neural networks applied to gridded crime surfaces capture spatial patterns at multiple scales (Wang et al., 2019). Recurrent architectures, particularly LSTMs and BiLSTMs, have been applied to crime time-series prediction to capture temporal dependencies in weekly and monthly crime trends (Butt et al., 2024). Graph neural network approaches model crime propagation across spatial networks, capturing dependencies between adjacent zones that grid-based models cannot represent. Recent studies using graph attention and spatio-temporal graph architectures have reported improvements over non-spatial baselines in crime forecasting tasks by incorporating both spatial adjacency and temporal lag structures (Huang et al., 2022).

More recent work has explored transformer-based spatio-temporal architectures capable of modelling long-range temporal dependencies and spatial interactions within crime forecasting problems. These models apply attention mechanisms to capture complex temporal relationships beyond the capabilities of standard recurrent architectures. At the same time, there has been growing interest in integrating auxiliary contextual information into crime forecasting frameworks. Recent crime forecasting studies have incorporated auxiliary contextual data such as human mobility, social media, taxi-flow, and weather variables, reporting improvements over models based only on historical crime records (Wu et al., 2022; Salama et al., 2021). These developments reflect a broader shift toward increasingly data-rich and architecturally flexible forecasting systems.

Despite these advances, the fundamental output of most deep learning crime forecasting models remains a predicted count or occurrence probability over a fixed future time window. This is a meaningfully different quantity from the recurrence timing that survival models produce. Two police beats predicted to experience similar counts over the next month may differ substantially in whether that activity is likely to occur in the next three days or the next three weeks, a distinction that is critical for patrol scheduling but largely invisible to count-based forecast models. Deep learning count models therefore do not naturally provide the timing resolution required to rank areas according to proximity to their next event. This distinction is illustrated in Figure 1, which contrasts count-based forecasts with recurrence timing predictions for two beats with similar expected counts but different temporal profiles.

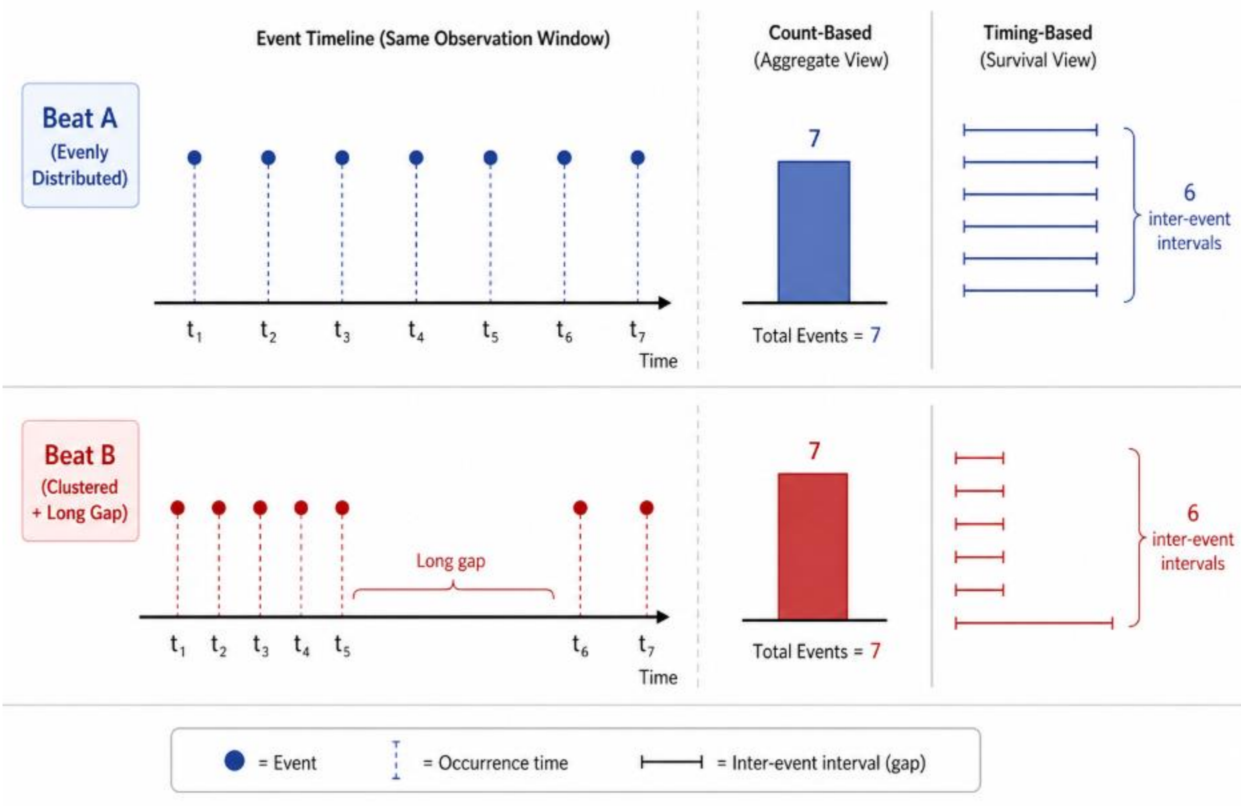

Figure 1: *Comparison of count-based prediction and recurrence timing. Similar predicted counts can mask substantial differences in when events are expected to occur*

Furthermore, growing critical attention has been paid to the evaluation rigour of crime prediction models. Prior work shows that predictive performance can be sensitive to grid design, spatial aggregation, evaluation windows, and the structure of the test data (Roberts et al., 2017; Rosser and Cheng, 2017; Kounadi et al., 2020). This concern reinforces the importance of the time-aware evaluation framework adopted in the present study.

A further practical concern is data requirements. Graph convolutional and transformer-based architectures require large numbers of spatial units with dense event histories to learn meaningful spatial representations. In settings with fewer geographic units such as the police beat level, these models may not have sufficient training signal to outperform simpler approaches, a possibility that motivates the comparative evaluation framework here.

2.5 Deep Learning for Survival Analysis

The intersection of deep learning and survival analysis has produced a rapidly growing body of methodological work. DeepSurv (Katzman et al., 2018) replaces the linear predictor of the Cox model with a feed-forward neural network, allowing non-linear covariate effects while retaining the partial likelihood estimation framework. DeepHit (Lee et al., 2018) reformulates survival prediction as discrete-time classification, avoiding the proportional hazards assumption entirely. Dynamic-DeepHit (Lee et al., 2020) extends this to time-varying covariates, enabling predictions to update as new observations arrive, a capacity that is conceptually attractive for crime recurrence, where the most recent inter-event gap may be the strongest predictor of the next.

The field has evolved substantially since these foundational models. Kvamme et al. (2019) provide an extensive benchmark comparison of neural survival models, finding that performance gains over Cox are real but inconsistent, and that they are most pronounced when covariate effects are genuinely non-linear and sample sizes are large. Random survival forests (Ishwaran et al., 2008) continue to serve as a competitive non-parametric baseline, often outperforming deep models on moderate-sized datasets. Wang and Sun (2021) propose a transformer-based survival model, SurvTRACE, that leverages attention mechanisms to capture complex interactions between features, reporting state-of-the-art performance on several medical benchmarks. More recent work has also explored deep survival models capable of incorporating longitudinal and time-varying covariates directly into the prediction process. Nagpal, Jeanselme and Dubrawski (2021) demonstrate that neural survival frameworks can integrate evolving temporal information, allowing survival predictions to update dynamically as new observations become available. Whether this additional complexity translates into meaningful improvements over simpler survival models in the crime domain, where per-unit event histories are sparser and feature sets less rich than in clinical data, remains an empirical question that this study is well positioned to examine.

Wiegrebe et al. (2024) provide a recent comprehensive review of deep survival analysis, identifying the main architectural families, benchmark datasets, and open challenges. They show that conclusions about model superiority depend strongly on model class, data structure, and evaluation metric. For the present study, this distinction is important because the evaluation focuses on how well models distinguish shorter from longer robbery recurrence intervals using the C-index and time-dependent AUC. Calibration was not evaluated because the aim was not to estimate exact recurrence probabilities at fixed horizons, such as the probability of another robbery within 7 or 30 days. Evaluating that would require additional probability-based assessment, such as Brier scores or calibration plots. Therefore, the findings should be interpreted as evidence that the models can compare recurrence timing between observations, not as evidence that they produce fully calibrated probability forecasts.

2.6 Synthesis and Identified Gap

Across the literature, three important strands can be identified. First, place-based criminology and near-repeat research show that crime is concentrated in particular locations and that recent events can carry information about future risk (Sherman, Gartin and Buerger, 1989; Townsley, Homel and Chaseling, 2003; Johnson, 2008; Weisburd, 2015). Second, survival analysis provides a direct framework for modelling event timing, but criminological applications have focused mainly on individual-level outcomes such as recidivism rather than area-level crime recurrence. Third, self-exciting and spatio-temporal prediction models capture event dependence and spatial clustering, but they are often framed around intensity surfaces, hotspots, or fixed-window counts rather than the duration between consecutive incidents within the same policing unit.

The gap is therefore not simply the absence of crime prediction models, but the absence of a survival-based framework that treats geographic areas as recurrent units at risk. Existing approaches provide useful insights into where crime concentrates and how recent events shape risk, but less attention has been given to modelling how long a police Beat remains without another robbery after a current event. This is important because recurrence timing is a different problem from count prediction or hotspot detection.

This study addresses that gap by framing robbery recurrence at the police Beat level as an inter-event survival problem. It constructs interval-level observations between consecutive robberies, engineers temporally valid features representing recurrence history, neighbourhood activity, seasonality, and self-excitation, and compares Cox, DeepSurv, LSTM-Cox, and self-excitation-enhanced Cox models under a time-aware evaluation framework. This allows the study to examine not only which model performs best, but also whether predictive improvement comes from model complexity or from theoretically grounded feature design.

The study also responds to concerns about evaluation rigour in crime prediction. Predictive models in this field can produce misleading results if future information enters training or if evaluation ignores chronological ordering (Perry et al., 2013). By using a time-aware train–validation–test split and evaluating discrimination through the C-index and time-dependent AUC, the study adopts an evaluation design aligned with the forward-looking task of modelling the waiting time to the next robbery.

Table 1: Summary of the major theoretical and methodological domains related to crime recurrence modelling, highlighting their core focus, limitations, and relevance to the present study

Theme Domain	Core Focus	Key Limitation in Existing Literature	Relevance to current Study
Place-based criminology	Spatial concentration and temporal stability of crime at micro-places	Strong on explaining where crime concentrates, weaker on modelling recurrence timing between events	Justifies beat-level modelling and the assumption that place histories contain predictive information
Near-repeat victimisation	Elevated short-term risk following prior events	Primarily descriptive; limited direct modelling of time-to-next-event at specific locations	Motivates inter-event survival framing and temporal decay features
Survival analysis	Time-to-event and recurrent event modelling	Applied predominantly to individuals rather than geographic crime recurrence	Provides core methodological framework for modelling recurrence timing
Self-exciting point processes	Contagion and event-driven clustering through conditional intensity modelling	Models event intensity effectively, but is less directly aligned with Beat-level inter-event survival modelling	Closest competing framework; motivates comparison between intensity and survival-based approaches
Spatio-temporal crime prediction	Crime count forecasting using KDE, RTM, CNNs, GNNs, and transformers	Focuses on counts or hotspot intensity rather than timing of recurrence	Demonstrates strengths of modern forecasting while highlighting the

			absence of recurrence timing prediction
Deep survival	Non-linear and sequence-aware survival prediction	Limited application in criminology and uncertain gains over simpler survival models	Motivates evaluation of DeepSurv and sequence-based survival approaches in crime recurrence settings
Present study	Area-level recurrent crime survival modelling	Existing approaches rarely model crime recurrence directly as a time-to-event problem at the geographic unit level	Integrates temporal recurrence modelling, spatial context, and time-aware evaluation to predict time-to-next-event

CHAPTER 3: METHODOLOGY

3.1 Research Methodology

Crime rarely occurs in isolation; it unfolds as a sequence of events over time. Building on this, the analysis models recurrence by focusing on the waiting time from a current robbery event to the next robbery event within the same police Beat. The problem is therefore treated as a time-to-event task, where the objective is to model recurrence timing rather than simply predict whether crime occurs. This formulation allows the temporal dynamics of robbery recurrence to be captured directly within the modelling framework.

This study adopts a quantitative research approach because the research questions require measurable analysis of crime event timing, model performance, and feature effects. The dataset consists of timestamped robbery records, allowing recurrence intervals to be constructed numerically and evaluated through survival modelling. This approach is appropriate because it supports statistical comparison between models and performance evaluation using metrics such as the C-index and time-dependent AUC.

The analysis was conducted using recorded crime data from the Chicago Police Department. To ensure a coherent and interpretable modelling framework, the scope of the study was narrowed to robbery incidents, and recurrence was defined at the Beat level. This allows recurrence to be interpreted as a localised process rather than an aggregate citywide phenomenon. The methodological approach combines traditional statistical modelling with modern machine learning techniques. Four modelling approaches were implemented and compared:

- Cox Proportional Hazards model as a baseline survival model
- DeepSurv model to capture potential nonlinear relationships between features and hazard
- LSTM-Cox model to incorporate temporal dependencies through sequential event data
- Cox model with a self-excitation feature to capture decayed same-Beat event history

Feature engineering was used to represent temporal history, local robbery intensity, neighbourhood robbery activity, seasonality, and self-exciting recurrence patterns. In addition, a sequence-based representation of events was constructed for the LSTM-Cox model to capture dependencies across consecutive observations.

To ensure valid evaluation, a time-aware train–validation–test split was applied, where models were trained on historical data and evaluated on future observations. Survival outcomes were constructed within each split, with final Beat observations treated as right-censored. Model performance was assessed using the concordance index (C-index), which measures the ability to correctly rank event times. This approach allows for a structured comparison between baseline, nonlinear, sequence-based, and self-excitation-enhanced survival models, providing insight into the extent to which temporal dependencies and feature interactions improve the modelling of robbery recurrence timing.

3.2 Data Source and Structure

The dataset used in this study was drawn from the Chicago Police Department's "Crimes - 2001 to Present" dataset, published through the City of Chicago Data Portal (City of Chicago, 2025). The full dataset was downloaded on 28 March 2026 and contained 8,519,604 records across 22 columns, covering reported incidents from 1 January 2001 to 19 March 2026. Each record corresponds to a single reported crime event, providing event-level detail suitable for analysing temporal patterns and recurrence.

Each record contains multiple attributes describing the incident, including:

- Temporal information: date and time of occurrence (Date)
- Spatial identifiers: Beat, District, Ward, and Community Area
- Crime classification: Primary Type, Description, and FBI Code
- Contextual indicators: Arrest status and Domestic flag
- Geographic coordinates: Latitude and Longitude

The analytical extract was narrowed to robbery incidents. This produced 316,564 robbery records across 302 Beats, covering incidents from 1 January 2001 to 18 March 2026. After applying the minimum event-history threshold used to reduce sparse Beat sequences, 316,505 robbery records across 297 Beats remained. The threshold was applied to reduce unstable recurrence sequences in Beats with very limited robbery history. Official police Beat boundary matching was then applied for neighbourhood feature construction. Twenty-eight Beats did not have matching official polygon boundaries and were removed, affecting 12,724 robbery records, equivalent to 4.02% of the robbery dataset at that stage. Removal was preferred to estimating replacement centroids from robbery coordinates, as event-derived centroids could bias neighbourhood construction toward high-crime locations and distort the spatial representation of Beat proximity (Chainey and Ratcliffe, 2005). The final modelling dataset retained 303,781 robbery records across 269 police Beats.

The Chicago dataset was selected because it provides long-term, event-level records with Beat identifiers and precise timestamps, which are necessary for recurrence-based survival modelling. Alternative open datasets were considered but were less suitable for this task. UK Police street-level crime data provides approximate rather than exact offence locations, which limits its suitability for precise local recurrence modelling (Police.uk, n.d.). London crime data from the Metropolitan Police is commonly provided as monthly counts at borough, ward, or LSOA level, making it more aggregated than the event-level timestamped structure required for modelling consecutive robbery intervals (London Datastore, n.d.).

Figure 2 provides an overview of the distribution of major crime types recorded in Chicago over the study period. Theft and battery dominate the dataset, while robbery represents a smaller but still substantial share of total incidents.

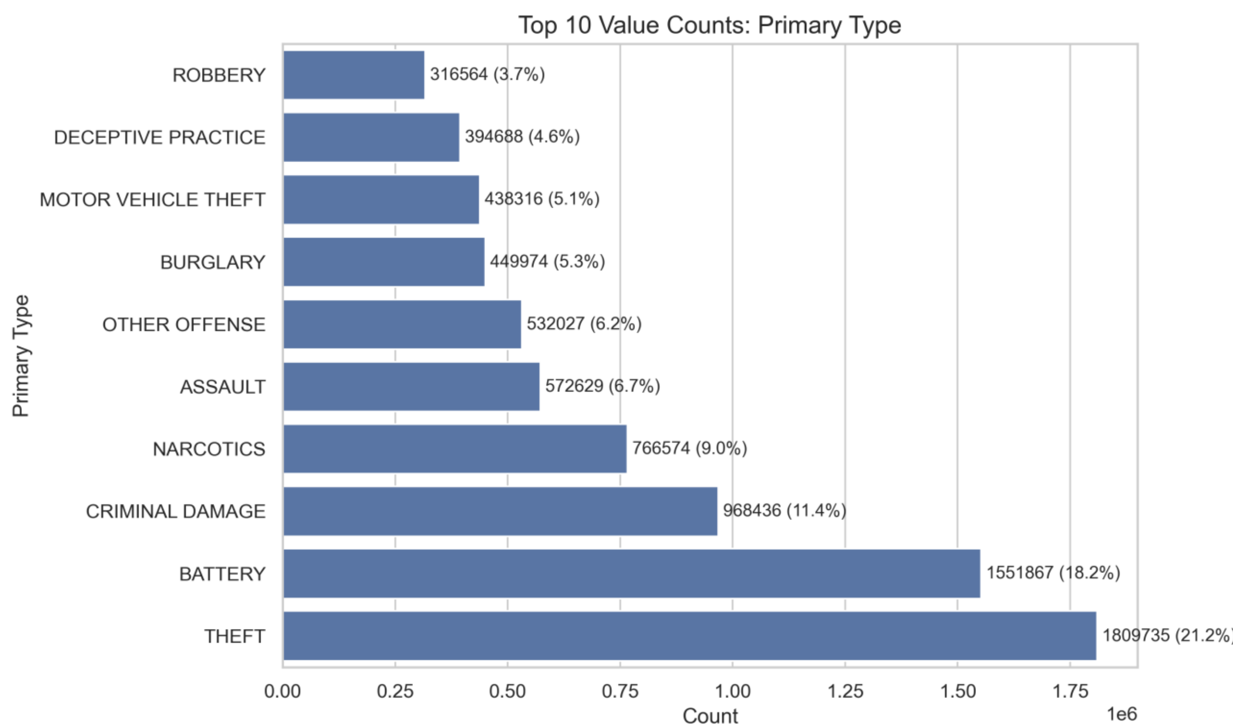

Figure 2: Distribution of major crime types recorded in Chicago over the study period (2001–2025). Source: Chicago Police Department

3.3 Data Preprocessing

All preprocessing steps were implemented in Python using pandas (McKinney, 2010), with GeoPandas and scikit-learn used for spatial and modelling preparation where required. The dataset was prepared to ensure consistent temporal ordering, reliable Beat-level grouping, and leakage-aware construction of recurrence intervals.

The Date field was converted to datetime format, and records with missing or invalid Date or Beat values were removed. The data was then filtered to robbery incidents and sorted chronologically within each Beat. This created an ordered event sequence for each Beat, which was required to calculate recurrence timing.

The cleaned robbery records were first organised at event level, where each row represented a single robbery incident. For survival modelling, these event-level records were then transformed into interval-level observations by measuring the waiting time from each current robbery to the next robbery in the same Beat. This interval was stored as next_crime_gap, with an event indicator used to distinguish observed recurrence from right-censored intervals.

3.4 Scope Definition and Unit of Analysis

3.4.1 Narrowing to Robbery

The analysis was restricted to robbery incidents to model a more coherent and interpretable temporal process. Exploratory analysis showed that when recurrence was defined across the broader crime dataset, inter-event times became highly compressed due to the continuous occurrence of different crime types. In many cases, the median gap between events was extremely small, often on the order of a few hours. While these intervals are statistically valid, they are of limited practical value, as they reflect overall crime volume rather than the recurrence of a specific behavioural pattern.

Figure 3 illustrates the distribution of inter-event gaps across all crime types, showing a strong concentration at very short intervals. In contrast, Figure 4 presents the distribution for robbery incidents only, where gaps are more dispersed and easier to interpret.

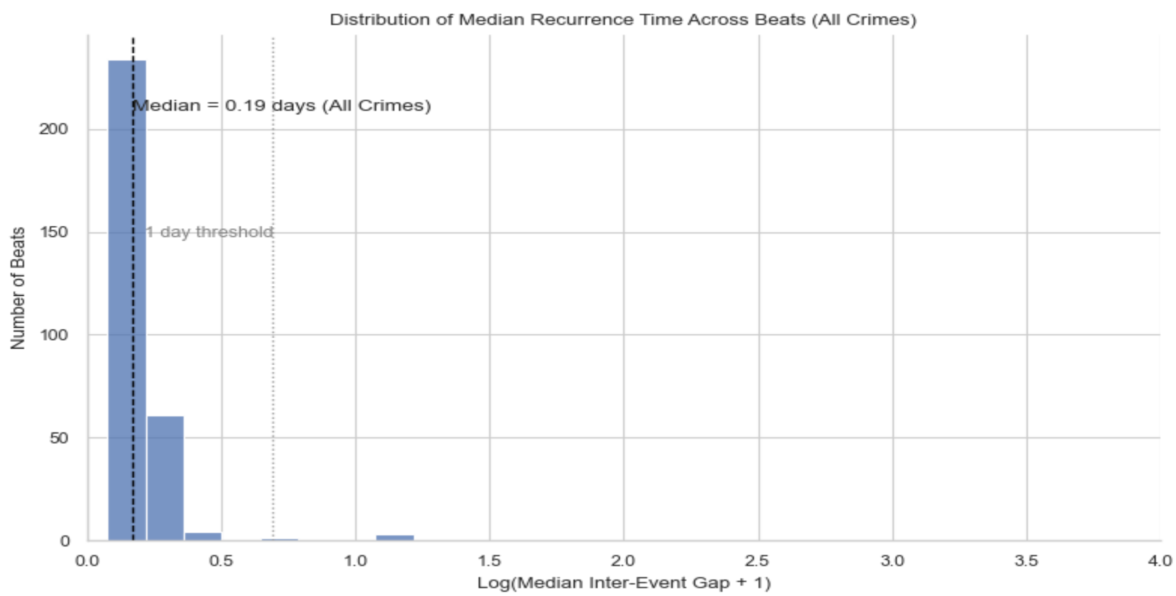

Figure 3: *Distribution of inter-event gaps across all crime types.*

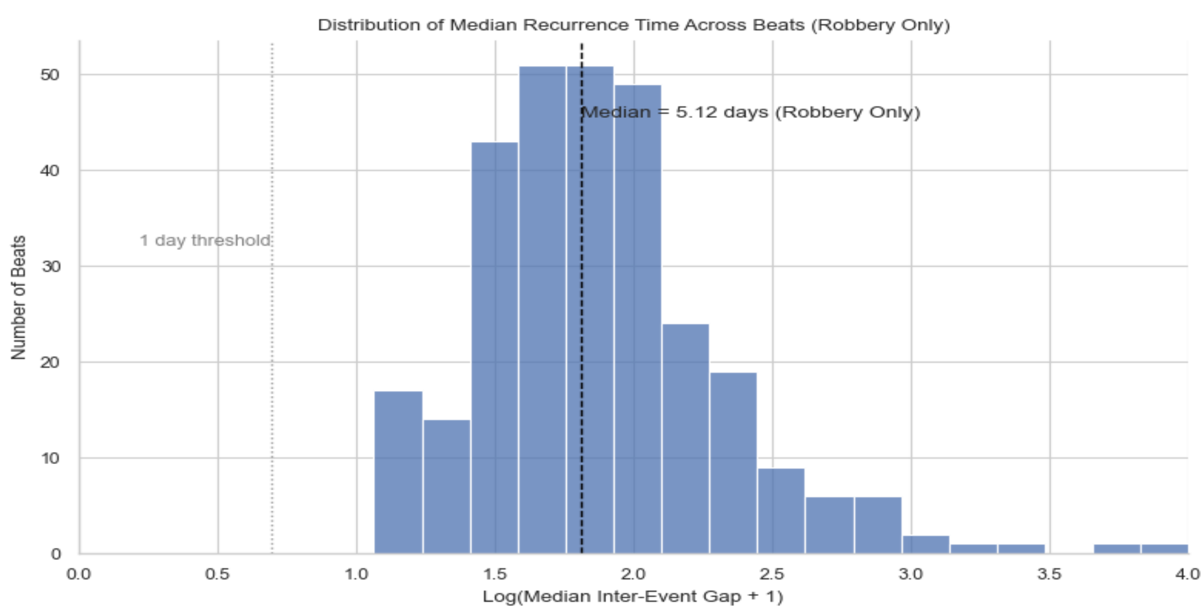

Figure 4: Distribution *of inter-event gaps for robbery incidents only.*

To address this, the analysis focuses on robbery as a single offence category. Compared to the aggregated crime stream, robbery produces inter-event intervals that are more interpretable and better reflect a consistent underlying process

3.4.2 Beat-Level Definition of Recurrence

Recurrence is defined as the waiting time from a current robbery event to the next robbery event within the same police Beat. The Beat is used as the unit of analysis because it provides a consistent spatial grouping for organising robbery events over time. By keeping recurrence within the same Beat, the resulting intervals can be interpreted as part of a local robbery process rather than a mixture of unrelated incidents across the city.

Other spatial units were considered. Larger areas, such as community areas, may provide more stable counts but can combine different local crime patterns, making recurrence timing harder to interpret. Smaller spatial units may provide finer detail but can produce sparse and fragmented event sequences. The police beat, therefore, provides a practical balance between local spatial detail and sufficient event history for survival modelling. This definition supports the modelling approach by allowing each Beat to be treated as a local sequence of robbery events, where recurrence timing is measured consistently from one event to the next.

3.4.3 Survival Formulation

The survival outcome was operationalised using two variables: next_crime_gap and event. The duration variable, next_crime_gap, measured the number of days between a robbery event and the next recorded robbery in the same Beat:

$$G_{i,b} = t_{i+1,b} - t_{i,b}$$

Where $G_{i,b}$ represents `next_crime_gap`, the waiting time from robbery event i to the next robbery in Beat b

The event indicator identified whether the interval ended with an observed robbery or was censored. Observations were coded as event = 1 when a subsequent robbery was observed within the relevant observation window. Where no subsequent robbery was observed before the end of the window, observations were coded as event = 0 and treated as right-censored.

Censoring was applied separately within the training, validation, and test periods. The final observed robbery in each Beat was therefore censored at the end date of the relevant split, ensuring that survival outcomes were constructed without using information from later evaluation periods.

3.5 Feature Engineering

Features were engineered to capture temporal recurrence, neighbourhood robbery activity, seasonality, and self-exciting event history. The feature design focused on representing both short-term recurrence patterns and longer-term Beat-level history, in line with the objective of modelling robbery recurrence timing. All predictors were constructed using information available at the current event or from earlier events, ensuring that future information was not used during feature construction. This was necessary to preserve the temporal integrity of the modelling process and reduce the risk of leakage in time-dependent prediction tasks (Hyndman and Athanasopoulos, 2021).

3.5.1 Temporal Recurrence Features

Temporal recurrence features were constructed from the ordered robbery sequence within each Beat. The variable prev_gap measured the number of days between the previous robbery and the current robbery, providing a direct measure of recent recurrence timing.

Rolling features were then created to summarise recent event history. avg_gap_3 and avg_gap_5 measured the average of the most recent three and five observed gaps within each Beat, while std_gap_5 captured variation in recent gap lengths. These features helped represent whether robbery recurrence was becoming more frequent, slowing down, or occurring irregularly.

Longer-term Beat history was captured using beat_total and days_since_start. beat_total measured the number of previous robberies in the same Beat before the current event, while days_since_start measured the time elapsed since the first recorded robbery in that Beat. For the first event in each Beat, previous-gap features were structurally missing because no earlier event existed. These cases were identified using a first-event indicator, with numerical missing values handled after the train–validation–test split using training-set values only.

3.5.2 Neighbourhood Activity

To capture local spatial context, a neighbourhood activity feature was constructed using proximity between police Beats. Neighbouring Beats were defined from official Chicago police Beat boundary polygons rather than event-based coordinates. The polygon geometries were projected into an appropriate metre-based coordinate reference system, and Beat centroids were calculated from these official boundaries. This avoided defining neighbourhoods from crime-event locations, which could shift centroids toward areas with more recorded robberies

A k-nearest neighbours approach was then applied, with k = 5, to identify the five nearest Beats for each Beat. Five neighbours were used to capture the immediate local context while limiting dilution from more distant Beats. Using these fixed neighbourhood relationships, neighbor_beat_count_last_30d was calculated as the number of robbery incidents recorded in neighbouring Beats during the 30 days before the current robbery event. This ensured that the feature captured recent nearby robbery activity using only information available before the prediction point.

3.5.3 Seasonality Features

To capture cyclical patterns in time, calendar variables were encoded using sine and cosine transformations. Month and day of week were represented as paired components (month_sin, month_cos, dow_sin, dow_cos), allowing cyclical relationships to be preserved.

This avoids the issue with direct numerical encoding, where adjacent periods can appear artificially distant (e.g., December and January). By mapping time onto a circular scale, the model can learn seasonal patterns more smoothly without discontinuities. These features allow the model to account for recurring temporal patterns, such as weekly and monthly variation in crime activity.

3.5.4 Self-Excitation Feature for Near-Repeat Effects

To capture short-term same-Beat event dependence, a self-excitation feature was constructed from previous robbery events within each Beat. For each current robbery, the feature summed the decayed influence of all earlier robberies in the same Beat. The current event was not included in its own calculation, ensuring that only past event history was used. Recent robberies therefore, contributed more strongly to the feature, while older robberies had progressively smaller influence. This reflects the near-repeat idea that robbery risk may be temporarily elevated after recent events in the same area (Johnson et al., 1997).

This feature was not treated as a full Hawkes process model. Instead, it was used as a lightweight self-excitation proxy within the survival modelling framework, allowing decayed same-Beat event history to be tested as an additional predictor of recurrence timing. This choice allowed event-driven temporal dependence to be tested within the survival framework without introducing the complexity of fitting a full point process model

3.5.5 Feature Transformation and Regularisation

Feature transformations were applied to reduce skewness and improve model stability. For the Cox and DeepSurv models, heavily skewed temporal and count-based variables were transformed using log(1+x), including previous gap, Beat history, days since Beat start, neighbourhood activity, standard deviation of recent gaps, and self-excitation.

Structural missing values in recurrence-history features, such as the first event in each Beat having no previous gap, were retained during feature construction and handled after the train–validation–test split. Numerical missing values were imputed using training-set values only, then applied consistently to validation and test data. This avoided using information from validation or test data during preprocessing. For neural models, feature scaling was also fitted on the training data only and then applied to validation and test sets.

3.6 Time-Aware Split and Leakage Control

A time-aware splitting strategy was used to preserve chronological order and support realistic model evaluation. Random partitioning was avoided because it can introduce temporal leakage by allowing future observations to influence training (Roberts et al., 2017).

The dataset was split sequentially. Training data included robbery events before 2018, validation data covered 2018–2020, and test data included events from 2021 onwards. Survival outcomes were constructed within each split, so the final observed robbery in each Beat was treated as censored at the relevant split end date. This ensured that training outcomes were not constructed using validation or test-period events.

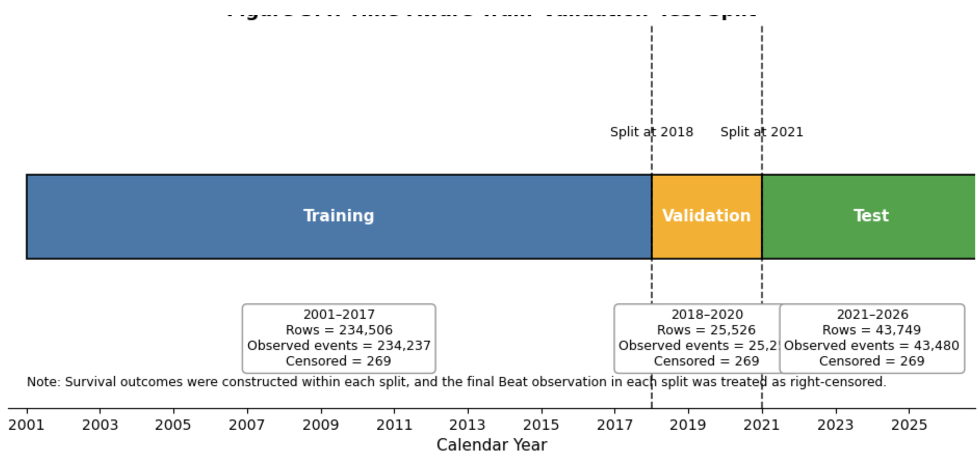

Figure 5: Time-aware train–validation–test split.

For the LSTM-Cox model, sequences were assigned to training, validation, or test sets using the timestamp of the target event. Validation and test sequences were allowed to include earlier historical events from the same Beat where available, but no sequence included events occurring after its target date. Feature scaling for neural models was fitted on the training set only and then applied to validation and test data.

3.7 Methodological Limitations

The analysis was conducted at the police Beat level. While this provides a consistent spatial unit for recurrence modelling, it may overlook finer within-Beat variation. Neighbourhood relationships were simplified using the five nearest official Beat centroids, which captures local proximity but does not fully represent road networks, physical barriers, or more complex spatial interactions.

Temporal dependence was represented through engineered recurrence features and a self-excitation proxy rather than a fully specified point process model. Therefore, the analysis identifies predictive patterns in robbery recurrence timing but does not claim to model the full underlying crime-generating process.

3.8 Modelling

All models were implemented in Python to ensure reproducibility. Survival modelling was conducted using lifelines (Davidson-Pilon, 2019), while DeepSurv was implemented using TensorFlow/Keras (Abadi et al., 2016; Chollet, 2015). The LSTM-Cox model was implemented using PyTorch (Paszke et al., 2019), which was used to construct sequence-based survival models. Data preprocessing, scaling, and evaluation utilities were handled using scikit-learn

(Pedregosa et al., 2011), while data manipulation and numerical operations were performed using pandas (McKinney, 2010) and NumPy (Harris et al., 2020).

3.8.1 Penalised Cox Proportional Hazards Model

The Cox Proportional Hazards model was used as the baseline for modelling robbery recurrence timing because it is interpretable and suited to time-to-event outcomes. The model estimates the hazard of the next robbery as:

$$h(t|x) = \ h_{0(t)} \exp(\beta^T \ x)$$

where $h(t|x)$ is the hazard at time t for a given set of predictors, $h_{0(t)}$ is the baseline hazard, and $\beta^T \ x$ is the linear predictor.

This formulation is appropriate because it does not require a fixed parametric distribution for recurrence times. Crime recurrence intervals are irregular, so imposing a specific hazard shape, such as exponential or Weibull, may be restrictive. Cox also provides a clearer survival framework than standard regression or classification approaches, which do not naturally handle recurrence timing or censoring (Cox, 1972; Kalbfleisch and Prentice, 2002).

The Cox model was treated as an interpretable baseline rather than a final modelling solution, as it assumes log-linear covariate effects and proportional hazards. These assumptions may not fully capture nonlinear relationships or changing temporal dependence in robbery recurrence, motivating comparison with more flexible survival models.

A selective feature set was used to reduce redundancy and improve stability. The final Cox specification included log-transformed recurrence and activity variables, including previous gap, recent average gap, recent gap variability, prior Beat activity, days since Beat start, and neighbourhood robbery activity. Seasonal effects were represented using cyclical month and day-of-week encodings, while previous arrest status, weekend timing, and first-event status were included as additional contextual indicators.

Not all engineered variables were retained. Several features captured overlapping aspects of recurrence history, which could increase multicollinearity and make coefficient estimates less stable. A selective specification was therefore preferred over exhaustive feature inclusion.

To further improve stability, L2 penalisation was applied. Penaliser values between 0.01 and 1.0 were evaluated using validation performance, with 0.5 selected for the final model. This provided a balance between reducing coefficient instability and avoiding excessive shrinkage.

3.8.2 DeepSurv Model

DeepSurv was implemented as a nonlinear extension of the Cox model. Instead of estimating risk through a linear predictor, DeepSurv uses a feedforward neural network to learn the risk function:

$$h(t|x) = h_{0(t)} \exp\big(f(x)\big)$$

where $f(x)$ represents the risk score learned by the neural network. The model retains the Cox partial likelihood loss, allowing it to remain within the survival modelling framework while relaxing the Cox model's log-linear assumption (Katzman et al., 2018).

The same survival outcome was used as in the Cox model, with next_crime_gap as the duration variable and event as the censoring indicator. Input features were standardised before training, using scaling parameters fitted on the training set only and then applied to validation and test data. The final network consisted of two hidden dense layers with 128 and 64 units, both using ReLU activation. L2 regularisation was applied to both hidden layers, with penalty values of 0.0001 and 0.001 respectively. Dropout of 0.1 was applied after each hidden layer, followed by a single output node representing the predicted risk score.

The model was trained using the Adam optimiser with a learning rate of 0.0003. Early stopping monitored validation loss with a patience of 12 epochs and restored the best-performing weights. Observations were sorted by descending survival duration before training, and full-batch training was used so that the Cox risk sets were constructed consistently within the partial likelihood loss.

3.8.3 Self-Excitation-Enhanced Cox Model

An additional Cox specification was implemented to test whether decayed same-Beat event history improved recurrence-risk ranking. This model used the same Cox framework and survival outcome as the baseline model but added log_self_excitation as an extra predictor.

The self-excitation feature represented the accumulated decayed influence of previous robberies within the same Beat. It was included to capture near-repeat temporal dependence while keeping the model interpretable and directly comparable to the baseline Cox model. This was not treated as a full Hawkes process; instead, it was used as a lightweight event-history feature within the Cox survival framework. Because this model retained the Cox structure, any performance difference could be interpreted as the contribution of the self-excitation feature rather than a change in modelling objective.

3.8.4 LSTM-Cox Model

The LSTM-Cox model was implemented to test whether ordered event sequences improved recurrence-risk ranking beyond fixed feature-vector models. Unlike Cox and DeepSurv, which use one feature vector per observation, the LSTM processes a short sequence of recent robbery events within the same Beat. The final hidden state of the LSTM is passed through a fully connected layer to produce a scalar risk score, which is incorporated into the Cox partial likelihood loss.

The LSTM input was structured as:

$$X \in R^{N\times 5\times d}$$

where N is the number of sequences and d is the number of features per event.

Each input sequence contained five consecutive robbery events. This window was chosen to capture recent event history while keeping the sequence short enough to avoid excessive sparsity and unnecessary model complexity. The sequence features included recurrence-history variables, cumulative Beat activity, neighbourhood robbery activity, seasonal encodings, previous arrest status, weekend timing, and first-event status. Features were standardised using training-set parameters before being applied to validation and test data.

The model was implemented in PyTorch using a single LSTM layer with hidden size 16, dropout of 0.1, and a fully connected output layer. A compact architecture was selected to reduce overfitting risk given the structured and relatively short event sequences. Adam optimisation was used with a learning rate of 0.0005 because it provides stable gradient-based optimisation for neural network training. Training was run for a maximum of 40 epochs, with early stopping patience set to 6 to stop training once validation performance stopped improving. Random seeds were fixed at 42 for NumPy and PyTorch to support reproducibility. The final LSTM-Cox model was selected from the epoch with the best validation C-index.

3.8.5 Cox Partial Likelihood Objective

All survival models were based on the Cox partial likelihood objective. The Cox and self-excitation Cox models used this objective through the lifelines implementation, while DeepSurv and LSTM-Cox used it as the training loss for neural risk scores.

The objective can be written as:

$$\mathcal{L}(\theta) = -\sum_{i:\delta_i=1}\left[r_i - \log\left(\sum_{j\in R_i(\mathrm{t_i})}\exp\left(r_j\right)\right)\right]$$

where $(\mathrm{r_i})$ is the predicted risk score for observation i, $(i\colon \delta_i = 1)$indicates that the recurrence event was observed, and R_i is the risk set containing observations still at risk at time $(\mathrm{t_i})$. Minimising this loss encourages the model to assign higher risk scores to observations that experience recurrence sooner.

This objective was used because the aim was to rank observations by recurrence timing rather than predict exact durations directly. The Cox partial likelihood compares relative risk among observations and does not require explicit estimation of the baseline hazard (Katzman et al., 2018; Kvamme, Borgan and Scheel, 2019).

For DeepSurv, observations were sorted by descending survival duration before training so that risk sets were constructed correctly in the custom loss function. For LSTM-Cox, risk scores produced from the final LSTM hidden state were passed into the Cox partial likelihood, with sorting handled during loss computation. This allowed the linear, nonlinear, sequence-based, and self-excitation-enhanced models to be compared under the same survival objective.

3.8.6 Evaluation and Benchmarking

Model performance was evaluated using the concordance index (C-index), which was applied consistently across all models. This allowed direct comparison between the Cox baseline, DeepSurv, LSTM-Cox, and the self-excitation-enhanced Cox model.

Validation performance was used for model selection, while test performance was reserved for final assessment. Training, validation, and test scores were compared to assess generalisation and identify possible overfitting, particularly in the neural survival models.

CHAPTER 4: Results

4.1 Experimental Setup

Four survival modelling approaches were evaluated: a Cox baseline, a self-excitation-enhanced Cox model, DeepSurv, and LSTM-Cox. Performance was assessed using the C-index under the time-aware train–validation–test split.

4.2 Distribution of Inter-Event Times

Figure 6 presents the distribution of inter-event times for robbery incidents in the final modelling dataset. While this distribution was introduced in Chapter 3 to motivate the focus on robbery, it is presented here as a formal characterisation of the outcome variable before survival analysis. The distribution is strongly right-skewed, with most gaps concentrated below 20 days and a long tail extending beyond 100 days. This indicates that robbery recurrence is rapid for the majority of beats, while a smaller proportion experience extended periods of inactivity.

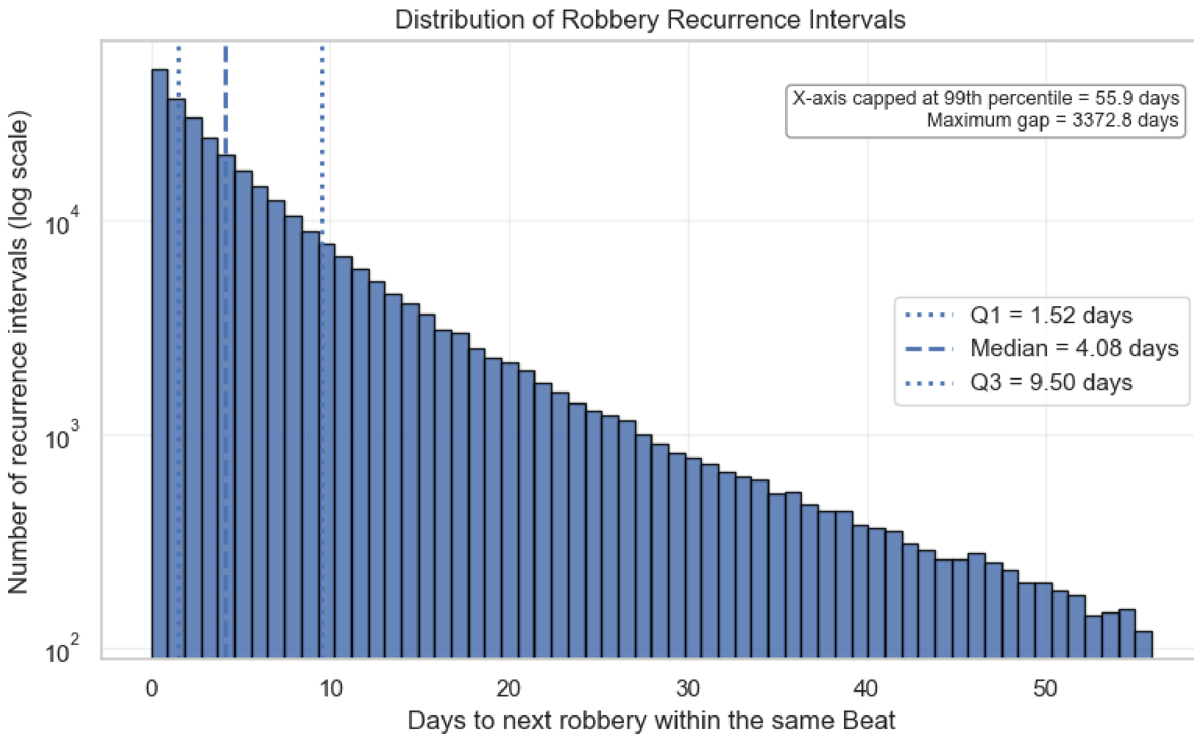

Figure 6: Distribution of robbery recurrence intervals in the final modelling dataset

The histogram shows that robbery recurrence intervals are strongly right-skewed. Most current-to-next robbery intervals are short, with a median of approximately 4.1 days and an interquartile range of about 1.5 to 9.5 days. This indicates that many Beats experience another recorded robbery within a relatively short period after a current event. However, the long upper tail, extending to over 3,000 days, shows that some Beats also experience much longer periods without recurrence. This uneven distribution supports the use of survival modelling, as the outcome is time-dependent, highly skewed, and not well represented by a simple average interval.

4.2.1 Kaplan–Meier Analysis

The Kaplan–Meier curve shows that robbery recurrence occurs quickly for a large share of observations. The median survival time is 4.09 days, meaning that half of observed recurrence intervals had experienced a subsequent robbery within just over four days. The survival probability falls to 0.338 by 7 days, 0.155 by 14 days, 0.043 by 30 days, and 0.009 by 60 days. This confirms that recurrence is concentrated in the short term, although a small number of intervals remain event-free for much longer.

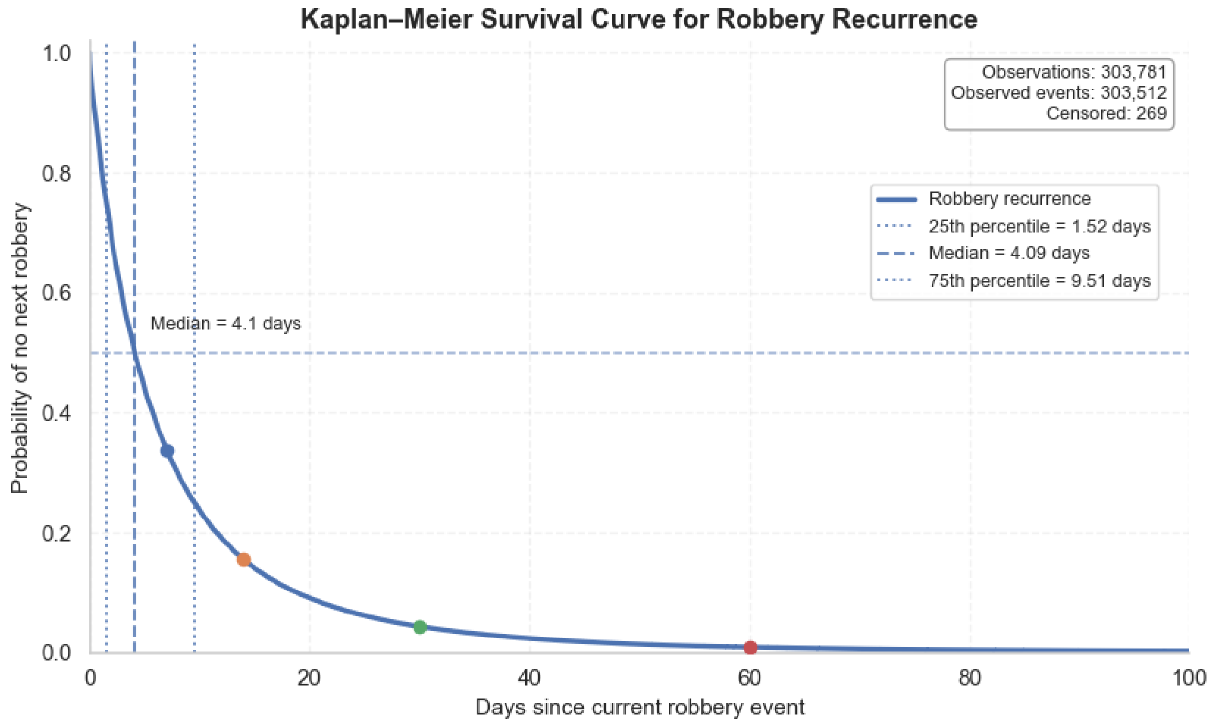

Figure 7: Kaplan*–Meier survival curve for inter-event times.*

4.3 Overall Survival Model Performance

Table 2 compares the performance of the four survival modelling approaches using the time-aware train–validation–test split. The C-index was used to evaluate how well each model ranked robbery events by recurrence timing.

Table 2: Overall survival model performance using C-index

Model	Train C-index	Validation C-index	Test C-index
Cox baseline	0.6124	0.6040	0.6179
Cox + self-excitation	0.6209	0.6131	0.6251
DeepSurv	0.6159	0.6051	0.6165
LSTM-Cox	0.5997	0.5956	0.6088

The Cox + self-excitation model produced the highest C-index across all three data splits, with a test C-index of 0.6251. The Cox baseline achieved a test C-index of 0.6179, while DeepSurv produced a similar test score of 0.6165. The LSTM-Cox model recorded the lowest test C-index among the four models, at 0.6088.

Across all models, validation C-index values ranged from 0.5956 to 0.6131, while test C-index values ranged from 0.6088 to 0.6251. The gap between training and test performance was small across the models, indicating broadly consistent performance across the time-aware splits.

4.4 Feature-Group Ablation Analysis

The table below presents the Cox-based feature-group ablation results. Feature groups were added progressively while keeping the survival modelling framework fixed, allowing the contribution of each feature group to be assessed through changes in C-index.

Table 3: Feature-group ablation results using Cox survival models

Feature set	Features	Train C-index	Validation C-index	Test C-index	Validation gain	Test gain
Static / aggregate only	4	0.6072	0.5834	0.5637	—	—
Temporal recurrence	7	0.6034	0.5973	0.6100	0.0140	0.0463
Neighbourhood activity	8	0.6118	0.6034	0.6174	0.0061	0.0074
Seasonality	13	0.6124	0.6041	0.6180	0.0007	0.0005
Self-excitation / full model	14	0.6209	0.6131	0.6251	0.0090	0.0072

The largest improvement occurred when temporal recurrence features were added to the static/aggregate feature set. Test C-index increased from 0.5637 to 0.6100, showing that recent recurrence history contributed substantially to recurrence-risk ranking. Adding neighbourhood activity further improved test performance to 0.6174, while seasonality produced only a small additional gain.

The full model, which included self-excitation, achieved the highest validation and test C-index values. Test performance increased to 0.6251, with a further gain of 0.0072 over the seasonality model. This shows that decayed same-Beat event history added predictive value beyond static, temporal, neighbourhood, and seasonal features.

4.5 Interpretation of the Self-Excitation-Enhanced Cox Model

Following the overall model comparison and ablation analysis, the self-excitation-enhanced Cox model was selected for interpretation because it achieved the strongest validation and test C-index.

Table 4: Hazard ratio interpretation for self-excitation-enhanced Cox model

Feature	**Hazard ratio**	**Interpretation**
log_self_excitation	1.354	Higher self-excitation is associated with faster robbery recurrence
log_neighbor_beat_count_last_30d	1.195	More recent nearby robbery activity is associated with faster recurrence
log_beat_total	1.043	Greater prior Beat robbery history is associated with faster recurrence
log_prev_gap	0.969	Longer previous gap is associated with slower recurrence
log_avg_gap_3	0.917	Longer recent average gaps are associated with slower recurrence
log_std_gap_5	0.905	Greater recent gap variability is associated with lower hazard
log_days_since_start	0.942	Longer Beat history duration is associated with lower hazard
prev_arrest	0.977	Previous arrest is weakly associated with lower recurrence hazard
is_weekend	0.994	Very weak / not statistically meaningful
is_first_event_in_beat	0.931	Not statistically meaningful here

P-values are not emphasised in the interpretation because the sample size is very large, meaning even very small effects can become statistically significant. The interpretation therefore focuses on hazard ratio direction and magnitude, which are more informative for understanding the practical contribution of each predictor.

The hazard ratio results show that the strongest positive association with faster recurrence was log_self_excitation (HR = 1.354), followed by recent neighbouring robbery activity (HR = 1.195). This indicates that decayed same-Beat event history and nearby recent robbery activity were the most important positive predictors of shorter recurrence intervals. In contrast, longer previous gaps, higher recent average gaps, greater recent gap variability, and longer time since Beat start all had hazard ratios below one, indicating association with slower recurrence.

4.6 Risk Stratification Using the Best Model

Risk groups were created from the predicted risk scores of the self-excitation-enhanced Cox model on the test set. Observations were divided into low, medium, and high-risk groups using tertiles of predicted risk.

Table 5: Recurrence intervals by predicted risk group

Risk group	N	Observed events	Median gap days	Mean gap days
Low risk	14,583	14,338	10.98	18.62
Medium risk	14,583	14,563	6.35	10.23
High risk	14,583	14,579	3.40	5.92

The risk groups show clear separation in recurrence timing. The high-risk group had the shortest median recurrence interval at 3.40 days, compared with 6.35 days for the medium-risk group and 10.98 days for the low-risk group. Mean recurrence intervals followed the same pattern, decreasing from 18.62 days in the low-risk group to 5.92 days in the high-risk group.

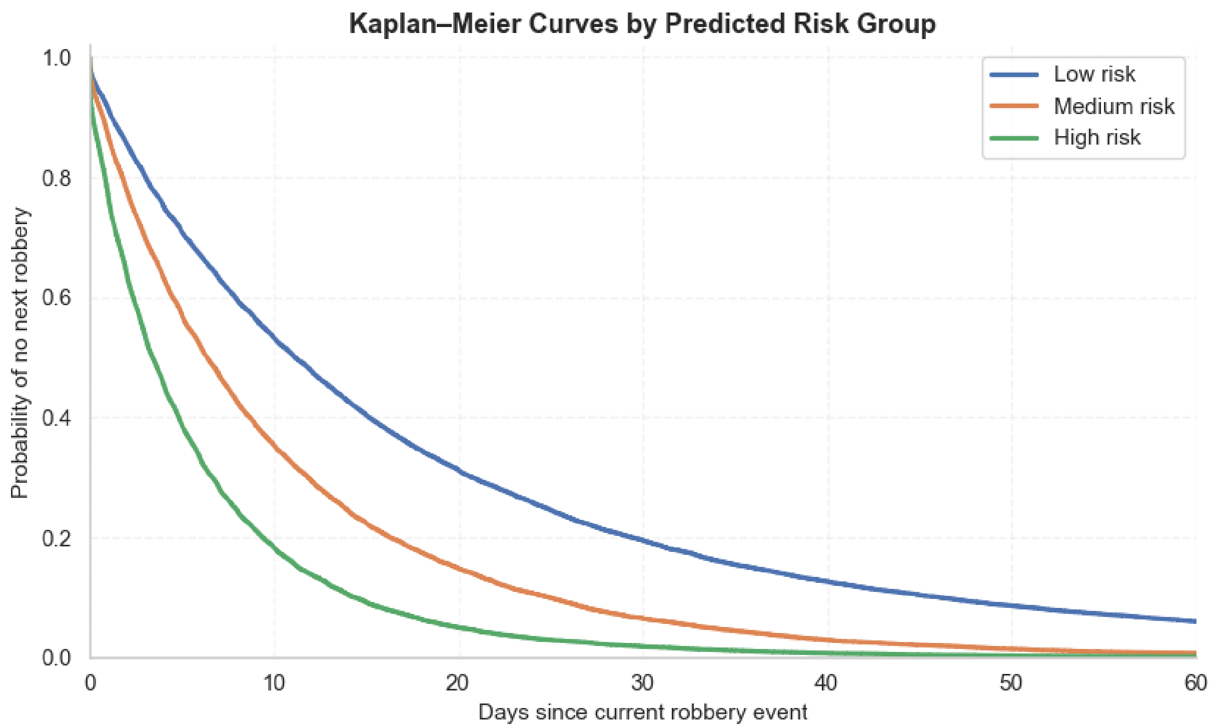

Figure 8: Kaplan–Meier survival curves by predicted risk group.

A multivariate log-rank test indicated that the survival curves differed across the predicted risk groups ($p < 0.001$). However, given the large test-set size, the practical interpretation is based mainly on the ordered recurrence intervals: the high-risk group had the shortest median gap of 3.40 days, followed by the medium-risk group at 6.35 days and the low-risk group at 10.98 days.

4.7 Time-Dependent AUC

Time-dependent AUC was used to assess how well the Cox + self-excitation model discriminated between shorter and longer recurrence intervals at different prediction horizons.

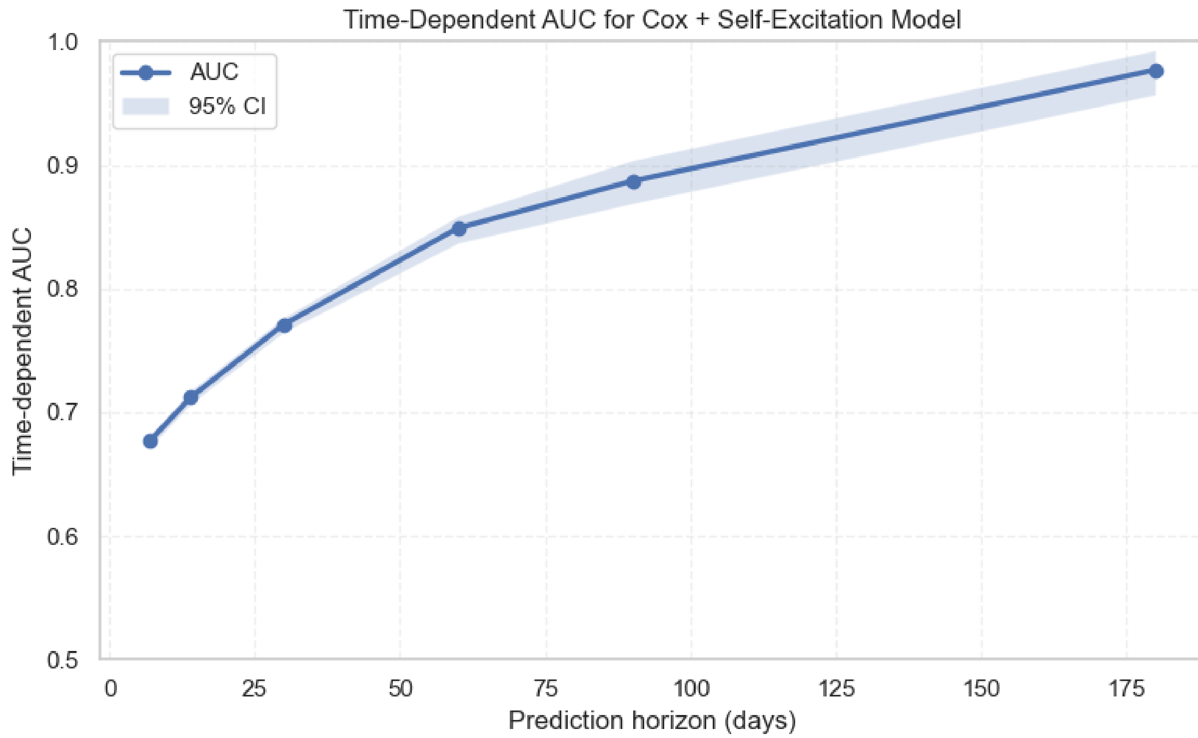

Figure 9: Time-dependent AUC for the Cox + self-excitation model with 95% confidence intervals.

Table 6: Time-dependent AUC results with 95% confidence intervals for the Cox + self-excitation model

Time horizon	**AUC**	**95% CI**
7 days	0.677	0.672–0.682
14 days	0.712	0.707–0.718
30 days	0.771	0.765–0.777
60 days	0.849	0.837–0.859
90 days	0.887	0.869–0.904
180 days	0.977	0.957–0.993

The AUC increased steadily across the prediction horizons, from 0.677 at 7 days to 0.977 at 180 days. The mean time-dependent AUC across the evaluated horizons was 0.717, with a 95% confidence interval of 0.712–0.721. This shows that discrimination was weaker at short horizons and stronger at longer horizons. The 7-day and 14-day values are most relevant for near-term recurrence prediction, while the high AUC at 90 and 180 days should be interpreted cautiously because many short recurrence intervals had already experienced the next robbery by those later time points. This makes separation between short- and long-duration observations easier at longer horizons.

4.8 Summary of Findings

Table 7: Summary of key findings by research question

Research Question	Key Finding
RQ1	Robbery recurrence intervals were strongly right-skewed and concentrated in short time windows. The median recurrence interval was 4.09 days, with survival probability falling to 0.338 by 7 days and 0.043 by 30 days.
RQ2	Survival models achieved moderate recurrence-risk ranking. The best-performing model was Cox + self-excitation, with a test C-index of 0.6251.
RQ3	Temporal recurrence features produced the largest improvement, increasing test C-index from 0.5637 to 0.6100. Neighbourhood activity and self-excitation added further gains, with the full model reaching 0.6251.
RQ4	DeepSurv performed close to the Cox baseline, while LSTM-Cox performed lower than the Cox-based models. The strongest performance came from the self-excitation-enhanced Cox model.
RQ5	Self-excitation showed the strongest positive association with faster recurrence, followed by recent neighbouring robbery activity. Longer previous gaps, longer recent average gaps, and longer time since Beat start were associated with slower recurrence.

CHAPTER 5: Discussion

5.1 Summary of Key Findings

This study examined robbery recurrence as a time-to-event problem at the Chicago police Beat level. Instead of modelling robbery only as counts or spatial concentration, the analysis focused on the waiting time from a current robbery to the next robbery within the same Beat. The results show that robbery recurrence is highly compressed in time. The median recurrence interval was 4.09 days, and the Kaplan–Meier curve showed that survival probability fell to 0.338 by seven days and 0.043 by 30 days. This confirms that recurrence is not only spatially uneven but also strongly time-dependent.

The strongest model was the self-excitation-enhanced Cox model, which achieved a test C-index of 0.6251. DeepSurv performed close to the Cox baseline, while LSTM-Cox performed lower than the Cox-based models. This suggests that model complexity alone did not improve recurrence-risk ranking. Instead, the largest gains came from feature design. Temporal recurrence features increased test C-index from 0.5637 to 0.6100, neighbourhood activity raised it further to 0.6174, and self-excitation produced the best full model.

The practical value of the model was clearest in the risk stratification results. The high-risk group had a median recurrence interval of 3.40 days, compared with 6.35 days for the medium-risk group and 10.98 days for the low-risk group. The findings therefore show that survival models can provide meaningful recurrence-risk ranking, but the central contribution comes from temporally valid recurrence-history features rather than black-box model complexity.

5.2 Interpretation of Results

5.2.1 Robbery Recurrence as a Short-Term Timing Process

The short median recurrence interval suggests that robbery recurrence at the Beat level is often a near-term process. Many Beats experienced another recorded robbery within days of the current event, while a smaller number had much longer quiet periods. This right-skewed structure supports the use of survival modelling because the outcome is not well represented by a simple average gap.

This finding is consistent with place-based crime research, which shows that crime concentrates strongly in a small number of locations (Sherman, Gartin and Buerger, 1989; Weisburd, 2015).

However, this study adds a timing dimension. It shows not only that crime is concentrated in places, but that recurrence within those places can happen quickly. This matters because hotspot mapping identifies where crime has concentrated, but it does not directly indicate how soon another event may occur.

The result also aligns with near-repeat and self-excitation perspectives. These theories suggest that a recent event can temporarily increase the likelihood of another event in the same or nearby area because opportunity, offender awareness, or environmental vulnerability may persist (Townsley, Homel and Chaseling, 2003; Johnson, 2008). The present results do not prove causal triggering, but they support the idea that recent robbery history contains useful information about near-term recurrence.

5.2.2 Model Performance and Predictive Limits

The model results show moderate but meaningful ranking ability. The best model, Cox + self-excitation, achieved a test C-index of 0.6251. This is above chance, but it is not high enough to support precise event prediction. The correct interpretation is that the model ranks recurrence risk better than random ordering, not that it can forecast the exact timing of the next robbery.

The risk stratification results make this more interpretable. High-risk observations had a median recurrence interval of 3.40 days, while low-risk observations had a median of 10.98 days. This shows that the model's risk scores translated into meaningful differences in observed recurrence timing. In operational terms, the model is more useful as a prioritisation tool than as a deterministic forecasting system. The time-dependent AUC results also need careful interpretation. AUC increased from 0.677 at seven days to 0.977 at 180 days, with a mean AUC of 0.717. The short-horizon values are the most relevant for near-term recurrence prediction. The very high long-horizon AUC should not be overstated because, by later time points, many short recurrence intervals have already failed. This makes separation between short- and long-duration observations easier. Therefore, the model appears stronger at long horizons, but the more demanding practical task remains short-term recurrence ranking.

The weaker performance of DeepSurv and LSTM-Cox suggests that the main predictive signal in this task was already captured by the engineered recurrence-history features. DeepSurv can learn nonlinear relationships, but it used the same structured feature set as the Cox baseline. If the strongest effects are relatively simple, such as shorter previous gaps, higher neighbourhood activity, and stronger self-excitation, then additional nonlinear flexibility may add little beyond what a regularised Cox model already captures. In this case, DeepSurv increased model flexibility without revealing a substantially stronger risk structure.

The lower performance of LSTM-Cox is also understandable. Although LSTM-Cox was designed to learn from ordered event sequences, much of the useful temporal information had already been summarised through features such as previous gap, rolling gaps, Beat history, neighbourhood activity, and self-excitation. This reduces the advantage of a sequence model. In addition, LSTM models have more parameters and can be more sensitive to noise, irregular event spacing, and temporal instability. Therefore, its weaker performance does not mean sequential information is irrelevant. Rather, it suggests that, in this dataset, simple temporally valid recurrence features captured the signal more reliably than a neural sequence architecture.

5.2.3 COVID-19 and Temporal Stability

The time-aware split should be interpreted in relation to COVID-19. The validation period covered 2018–2020, meaning it included the main disruption year, while the training period was pre-2018 and the test period began in 2021. This matters because COVID-19 altered mobility, routine activity, guardianship, and public interaction, all of which shape opportunity-based crime patterns. Recent studies found that pandemic restrictions changed urban crime patterns, with effects varying by crime type and city (Campedelli, Aziani and Favarin, 2021; Nivette et al., 2021).

This helps explain why validation performance was lower than test performance across all models. The validation set may have been harder to rank because it included an abnormal period in which robbery recurrence dynamics were disrupted. This should not be treated simply as random variation. It suggests that recurrence-risk models depend on temporal stability, and that major social shocks can weaken relationships learned from historical data.

This does not invalidate the modelling framework. Instead, it highlights a clear extension for future work. Recurrence models should be tested separately across pre-COVID, COVID-restriction, and post-restriction periods. That would allow the analysis to distinguish ordinary temporal generalisation from performance under structural disruption.

5.2.4 Contribution of Feature Groups

The feature ablation results show that feature design was the main source of improvement. The weakest model used only static and aggregate features, with a test C-index of 0.5637. Adding temporal recurrence features increased test performance to 0.6100, the largest gain in the ablation sequence. This shows that recent recurrence history is central to modelling time to next robbery.

Neighbourhood activity added further value, increasing test C-index to 0.6174. This suggests that recurrence timing is not fully contained within individual Beats. Recent robbery activity in nearby Beats also carries predictive information, which is consistent with near-repeat and spatial contagion ideas. However, the gain from neighbourhood activity was smaller than the gain from temporal recurrence features, meaning the strongest signal came from the Beat's own recent history.

Seasonality added very little, increasing test C-index only from 0.6174 to 0.6180. This does not mean that calendar patterns do not exist. Rather, it suggests that broad month and day-of-week patterns were less useful for ranking recurrence timing than recent event history. In this study, short-term recurrence pressure mattered more than broad seasonal structure.

Self-excitation produced the best full model, raising test C-index to 0.6251. This is important because the feature improved performance even after previous gaps, rolling recurrence patterns, neighbourhood activity, and seasonality were already included. It therefore captured more than the most recent gap. It represented accumulated event pressure within the same Beat, which appears to be a useful signal for recurrence-risk ranking.

5.2.5 Key Predictors of Shorter Recurrence Intervals

The hazard ratios reinforce the ablation findings. The strongest positive association was log_self_excitation, with a hazard ratio of 1.354. This indicates that higher decayed same-Beat event history was associated with faster recurrence. Recent neighbouring robbery activity also had a positive association, with a hazard ratio of 1.195. These were the clearest indicators that recent local and nearby event pressure mattered.

Several recurrence-history variables had hazard ratios below one. Longer previous gaps, longer recent average gaps, greater recent gap variability, and longer time since Beat start were associated with lower recurrence hazard. This is coherent: Beats with longer recent waiting times show weaker immediate recurrence pressure, while Beats with compressed recent histories show greater risk of faster recurrence.

These effects should be interpreted as predictive associations, not causal mechanisms. The self-excitation feature may reflect offender return behaviour, repeated vulnerability, persistent environmental opportunity, or unobserved reporting and policing patterns. The model cannot separate these mechanisms. What it does show is that decayed same-Beat event history improves recurrence-risk ranking in an interpretable survival framework. This connects the survival model to self-exciting point process work such as Mohler et al. (2011), while remaining simpler and easier to interpret than a full Hawkes model.

5.3 Theoretical Contributions

The study contributes by linking place-based criminology, near-repeat theory, and survival analysis. Place-based research explains why some locations remain persistently active. Near-repeat theory explains why recent events may temporarily elevate local risk. Survival analysis provides the timing framework for modelling how long a Beat remains without another recorded robbery. Combining these perspectives shifts the question from "where is robbery concentrated?" to "how soon is recurrence likely?"

The study also shows that self-excitation can be represented within a Cox survival framework. Full point process models are powerful but can be complex and less transparent. A decayed event-history feature offers a simpler way to test whether accumulated recent event pressure improves recurrence-risk ranking. The improvement in the self-excitation Cox model suggests that this middle-ground approach has value.

A further contribution is the finding that neural models did not outperform the best Cox-based model. DeepSurv and LSTM-Cox were included to test whether nonlinear or sequential architectures added predictive value. The results show that, in this setting, theoretically grounded features were more useful than increased model flexibility. This supports a critical position in applied modelling: complex methods are not automatically superior when the main signal can be represented through transparent, well-designed features.

5.4 Practical and Policy Implications

The findings have practical value for crime analysis, but they should be used cautiously. The model does not predict the exact time or location of the next robbery. It ranks observations by recurrence risk. This makes it more suitable for analytical prioritisation than automated operational deployment. The risk stratification results show why this matters. High-risk observations had a median recurrence interval of 3.40 days, while low-risk observations had a median interval of 10.98 days. This difference could help analysts identify Beats where recent robbery history suggests elevated short-term recurrence pressure. Unlike static hotspot maps, the survival framework adds a timing dimension.

However, the model should not be used for individual-level targeting or automatic enforcement. It operates at Beat level, but it is still based on recorded crime data. Recorded crime can reflect reporting behaviour, policing intensity, and institutional practices, not only underlying crime occurrence. Predictive policing research has shown that models trained on recorded crime data can reproduce historical biases if used without safeguards (Lum and Isaac, 2016; Richardson, Schultz and Crawford, 2019). Therefore, any practical use should involve transparency, human oversight, and regular auditing.

The moderate C-index also limits operational claims. A test C-index of 0.6251 is useful but not strong enough for high-stakes automated decisions. The model is best understood as a supplementary decision-support tool that adds temporal information to existing spatial analysis.

5.5 Limitations

Several limitations remain. First, the analysis uses officially recorded crime data. The model captures recorded robbery recurrence, not all robbery activity. Under-reporting, policing intensity, and recording practices may influence the observed patterns.

Second, the Beat is an administrative unit. It is operationally meaningful, but it can hide important street-level variation. Crime is often concentrated at micro-places, so Beat-level modelling may smooth over sharper local recurrence patterns.

Third, the neighbourhood feature is simplified. Five nearest Beat centroids provide a reproducible spatial definition, but they do not capture roads, barriers, land use, public transport, or offender movement. Spatial relationships in real cities are more complex than centroid proximity.

Fourth, self-excitation is used as a proxy rather than a full Hawkes process. It measures decayed same-Beat event history, but it does not estimate a full event intensity function or separate background risk from triggered risk.

Fifth, the study is predictive, not causal. Hazard ratios show associations with shorter or longer recurrence intervals. They do not show that a feature causes robbery recurrence. Causal claims would require a different design.

Finally, performance is moderate. The best C-index was 0.6251, which shows useful ranking but substantial uncertainty. The high long-horizon AUC also requires caution because later horizons are easier once many short gaps have already failed.

5.6 Future Research Directions

Future work should test this framework on other cities and crime types. Robbery has specific opportunity structures, so recurrence dynamics may differ for burglary, theft, assault, or vehicle crime. Replication would show whether Beat-level survival modelling generalises beyond Chicago robbery.

Future studies should also use smaller spatial units such as street segments, grid cells, or micro-places. This may reveal recurrence patterns hidden by Beat-level aggregation, although it would create additional sparsity challenges.

More dynamic models could also be developed. Rolling retraining, time-varying covariates, and regime-specific models could help account for temporal instability. The COVID-19 issue in this study shows why such extensions matter. Models should be tested across stable and disrupted periods rather than assuming that recurrence relationships remain constant.

Future research could also compare the Cox-based self-excitation proxy with full Hawkes models, neural point processes, or hybrid survival-point-process models. This would clarify whether formal event-intensity modelling adds value beyond the simpler interpretable approach used here.

Finally, future work should examine fairness and governance. Because recorded crime data can reflect unequal reporting and policing, recurrence-risk models need auditing before operational use. Future studies should assess whether risk rankings concentrate attention on already over-policed communities and how safeguards might reduce harm.

CHAPTER 6: Conclusion

This dissertation shows that robbery recurrence at the police Beat level can be modelled as a survival problem. Recurrence intervals were strongly concentrated in short time windows, with a median recurrence interval of 4.09 days. The self-excitation-enhanced Cox model achieved the strongest performance, with a test C-index of 0.6251. The main conclusion is that predictive improvement came more from temporally valid feature design than from neural model complexity. Temporal recurrence features produced the largest ablation gain, and self-excitation further improved performance beyond temporal, neighbourhood, and seasonal predictors. DeepSurv and LSTM-Cox did not outperform the best Cox-based model.

The study contributes a timing-based framework for analysing robbery recurrence and shows that recent event history, neighbourhood activity, and self-excitation can be incorporated into an interpretable survival model. It also shows the limits of prediction: performance is moderate, recorded crime data may reflect reporting and policing bias, and hazard ratios are predictive associations rather than causal effects. Overall, the strongest value of the work lies in transparent, theory-informed modelling of recurrence timing rather than black-box prediction.

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